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Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT When a load is restored or a line is reclosed, the electrical response can persist and resemble a new intervention. We formulate source attribution from eight phasor measurement units (PMUs) in the IEEE 39-bus system using switched differential-algebraic predictors. The bank carries the estimated state through restoration and retains the preceding intervention as a candidate explanation of the remaining response. Nominal signatures describe separation under the assumed plant and noise model. With thresholds calibrated separately to a target of one normal alarm per hour, the bank correctly attributed the family and source of 99/108 interventions before physical restoration, compared with 72/108 for a continuous generalized likelihood ratio observer and 22/108 for ExtraTrees. The bank gained 25.0 percentage points over the continuous observer (95% interval [13.89, 37.04] across 18 physical parents) and emitted 8 unmatched alarms, compared with 85 and 0, respectively. No method emitted a normal alarm in 3.92 monitored hours. At fixed thresholds in a separate control, resetting the estimated physical state reduced timely attribution from 108/108 to 75/108. Independent per-PMU noise at 0.071% added RMS total vector error degraded load attribution. The bank–comparator contrast does not isolate history retention. The results describe sequential, nonoverlapping interventions in a known catalog on one simulator family.

INDEX TERMS change detection and isolation, differential-algebraic equations, event attribution, measurement integrity, phasor measurement units, physics-informed inference, power system dynamics.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, physical state continues to evolve. A second intervention may begin before that response decays. With eight PMUs, the observed change can combine recovery from the earlier event with forcing from a new source (Fig. 1(a)).

The inference problem is to decide whether a new source is needed to explain the incoming reports. Because the preceding response is still evolving, the observation law depends on the earlier intervention. Classical change detection and isolation treats a single change of an otherwise stationary law [1]–[3], and the generalized-likelihood approach of Willsky and Jones detects jumps in a fixed linear system [4]. Power-system outage detectors compare network hypotheses under quasi-static or linearized models [5]–[7], and trajectory libraries diagnose faults from sparse PMUs [8]. These precedents motivate retaining a restored intervention as an explicit competitor when attributing later observations in a switched DAE network.

We develop a bank of switched-DAE predictors that carries the estimated state through restoration. Each history specifies a source, onset and restoration time; load severity is integrated analytically. Separate active and cause weights allow an intervention to end while its response remains an explanation of later reports (Section IV).

We then examine two consequences of this construction. Nominal signature energy describes source separation under the assumed plant and noise model, and a reset analysis quantifies the innovation bias caused by discarding physical state (Section V). A held-out comparison measures timely attribution and extra alarms against a continuous physical GLR and ExtraTrees after calibrating each procedure to the same normal-alarm target (one normal alarm/h; Section VII-A). This comparison evaluates complete procedures. Isolating the benefit of explicit recovery histories requires controls that distinguish state continuity from the uncertainty lost when histories are compressed.

The evaluation covers a known catalog of sequential load

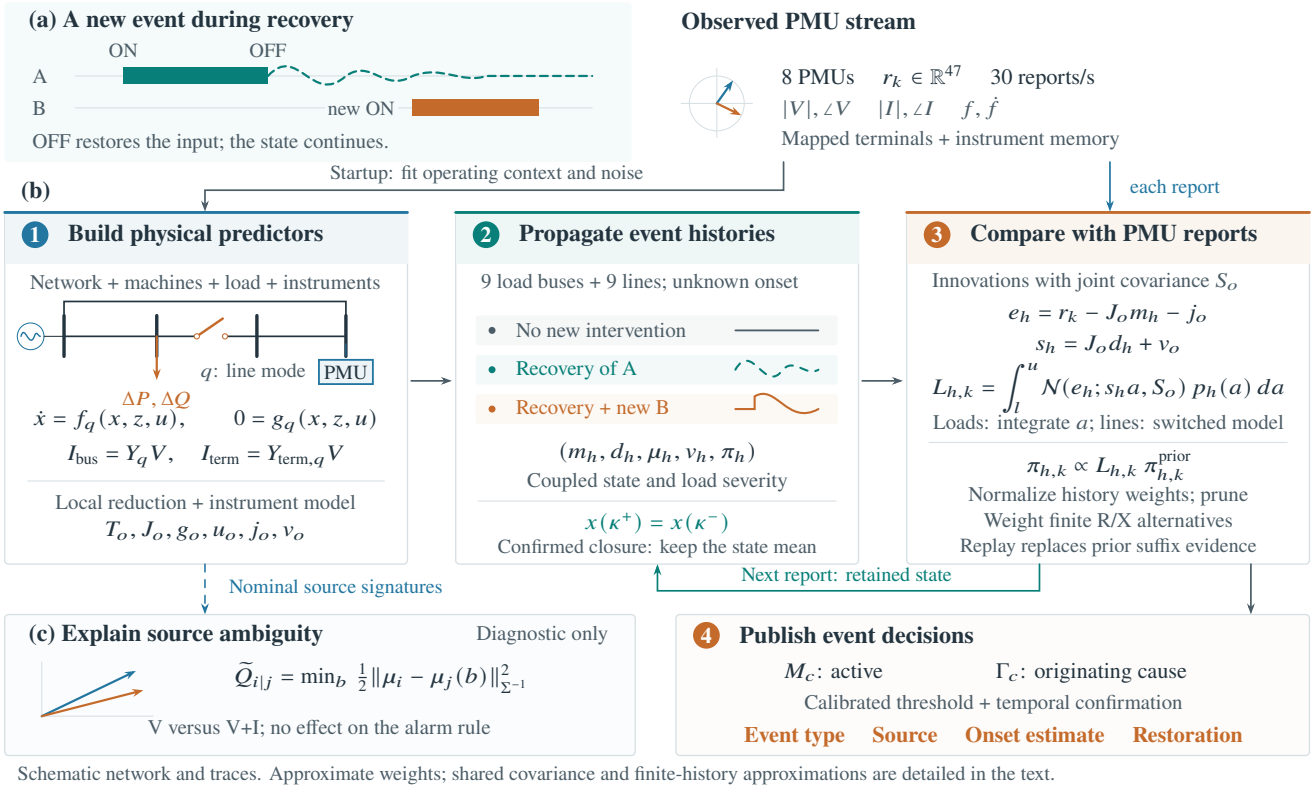


FIGURE 1. Recovery-aware event attribution from sparse PMUs. (a) A new intervention can begin while an earlier response persists. (b) Network-derived predictors compare recovery and new-event histories using the same report; integrated severity evidence updates their weights before detection, source diagnosis and restoration are published. The state continues through physical restoration; confirmed episode closure compresses histories to the retained state mean. (c) A separate nominal diagnostic measures source ambiguity; it does not gate publication. The electrical motif and traces are schematic.

and line interventions. The supplement records auxiliary analyses and reproducibility details.

II. RELATED WORK

A. CHANGE DETECTION AND ISOLATION IN DYNAMIC SYSTEMS

Sequential detection with a prior on the change time leads to the Shiryaev statistic, whose posterior-probability threshold is optimal for a geometric change-time prior [9]. Detection *and isolation* among several post-change laws was formalized by Nikiforov [1] and Lai [2]; asymptotically, the delay for a given error level is inversely proportional to the Kullback–Leibler information separating the true post-change law from its competitors [3]. For linear state-space systems, Willsky and Jones introduced signatures of jumps propagated through a Kalman filter and estimated their magnitude by generalized likelihood ratios [4]. Separating a constant bias from the state, as Friedland did for recursive filtering [10], gives the affine state–severity representation used in our load histories. Interacting and multiple-hypothesis filters propagate mode-conditioned states and mode-sequence histories with pruning and merging [11], [12]. Our method inherits these mechanisms. Its post-change models, however, are switched DAE operators of a power network, and a restored intervention remains a competing cause of later observations.

B. PHYSICAL INFERENCE OF GRID EVENTS

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [13]–[15]. Functional observability and descriptor-model diagnosability distinguish recovering a state quantity from identifying the input that produced it [16]–[19]. Phasor-angle hypotheses support outage identification [5], Bayesian outage inference with uncertain states [20], and quickest line-outage detection [6], [7]. Yang *et al.* add a time-varying AC likelihood chart [21] and particle-filter generator states [22] for persistent line outages, without restoration or loads. Anguluri *et al.* recover sparse forcing locations and inputs of linear dynamics by group LASSO [23], and sensor placement determines which sources are distinguishable [24]–[26]. Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window [8] and derive error/delay guarantees for an adaptive matrix CuSum [27]. Trajectory sensitivities and Gaussian evidence have been combined for parameter inference in power-system DAEs [28], [29]. PMU-based nodal-inertia estimation extends this neighboring line of work to spatially resolved dynamic parameters under ambient and transient conditions [30]. State and parameter estimates do not by themselves identify the intervention that produced them. Here the DAE-derived signatures provide a profiled separation diagnostic at a supplied onset, which we compare

retrospectively with attribution outcomes; it is not a general detectability guarantee. Le *et al.* recently proposed a physics-informed graph neural operator that maps sparse, intermittent PMU windows to unmeasured rotor angle and speed trajectories, with swing equations in the decoder and a learned correction for model mismatch [31]. That preprint addresses dynamic state reconstruction. Our method identifies which intervention explains observations while an earlier response is still evolving.

C. LEARNED DIAGNOSIS FROM SPARSE MEASUREMENTS

Learned and knowledge-driven representations identify PMU disturbances and assets [32]–[34]. For sparse-PMU transient-stability assessment, Wang *et al.* combine dominant-bus selection with a bidirectional recurrent assessor and an online check of model applicability under system changes [35]. This is a useful precedent for monitoring whether a learned assessment remains valid, but its output is a stability assessment rather than a source attribution for successive physical events. Barati's gated-recurrent-unit model predicts disruptions from historical synchrophasor streams, a forecasting task distinct from assigning an observed intervention to a physical source during recovery [36]. SHRED reconstructs IEEE-39 dynamic states from a few PMUs [37], and Dey *et al.* localize distribution topology changes with hardware-in-the-loop validation [38]. Graph neural networks encode the grid topology in the learned map, for event identification [39], physics-informed state estimation [40] and localization after reconstructing unmeasured trajectories [41]. These applications span state reconstruction and event diagnosis. In the present attribution task, the physical bank derives candidate responses from the network equations. Its signature analysis describes separation under the assumed plant and noise model. ExtraTrees provides a learned comparison on the same 18-source catalog.

D. MEASUREMENT INTEGRITY

Bad-data detection in state estimation tests residuals against the redundancy of a network model [42]; analytical-redundancy parity relations generalize this to dynamic systems [43]. For synchrophasors, robust estimators separate outliers from excursions [44], utility data show missing, locked, quantized and time-shifted reports [45], and attack identification characterizes which corrupted sensor sets remain distinguishable from system behavior [46]. Recent P-class synchrophasor-estimation work optimizes front-end accuracy and latency under IEC/IEEE measurement requirements [47], [48]; a review of PLL-based synchronization also relates front-end phase tracking to PMU applications in distributed generation [49]. End-to-end PMU tests under fault transients quantify how estimator errors affect a downstream application [50]. Multisnapshot state estimation models time-correlated instrument-transformer errors together with line-parameter uncertainty [51]; IEC 61850-9-2 co-simulation has also been used to assess OpenPMU and virtual-PMU state estimation [52]. These contributions address measurement generation, measurement uncertainty or state estimation. Our bank starts from reported phasor channels and does not validate a PMU estimator or a hardware-in-the-loop interface. The two parity relations used in the supplement require no state estimator and are connected to the information available for physical attribution when a PMU is lost.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the reference record released with the SGSMA 2026 competition. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The causal attribution test addresses load changes and line openings with restoration; missing and corrupted data are diagnosed by the integrity relations of the supplementary integrity analysis. Faults and generation changes provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the definition of absolute frequency. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

TABLE 1. Closest inference formulations. Only the quickest-change detector of Chen *et al.* and the two ExtraTrees adapters were executed on the common evaluation data.

Route	Mechanism	Evaluated target	Role in this study
Willsky–Jones GLR; Nikiforov/Lai	Filter signatures; detection–isolation statistics	One jump or one change in a fixed system	Information analysis (Section V)
IMM / M3H	Mode-conditioned filters, history pruning and merging	Hybrid switching-state estimation	History-bank mechanism
Chen / Rovatsos	Network likelihoods, sequential tests	Line outages	Chen executed on persistent openings
Yang	Time-varying AC likelihood chart; particle-filter states with EWMA	Persistent line outages, ten PMUs	Closest dynamic precedent, not executed
Warner high-fidelity	Simulated V/I paths, onset search	Short-circuit line candidates	Closest trajectory precedent
DAE / stochastic-state estimation	Physical state and parameter models	States, inputs or parameters	Components
Wang et al. sparse-PMU TSA	Bus selection; recurrent assessor; validity check	Stability assessment under grid changes	Adjacent; no source attribution
Le et al. GNO (arXiv 2026)	Graph attention; swing-equation decoder	Machine-state trajectories from sparse windows	Adjacent state estimation; not event attribution
Learned PMU diagnosis	Features or learned representations	Event families and locations	Two ExtraTrees adapters executed
This work	Switched-DAE histories with recovery	Successive catalog load/line events	Attribution, alarm cost, nominal separation

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Figure 4 maps these physical and measurement operations to the report consumed by inference. Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is generated by the noise model described in Section VI.

An intervention history is the discrete tuple $h = (e, c, \tau, \kappa, \mathcal{I})$: family, source, onset, restoration time ($\kappa = \infty$ for a sustained event) and, for a load, a severity interval $\mathcal{I}$. The severity $a \in \mathcal{I}$ is a continuous latent variable conditioned on h and integrated in the evidence; a line opening has no severity. A load changes its local P/Q setpoints by the common factor $1 + a$; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Parity diagnosis (supplement)
Unknown	Unresolved diagnosis	Not a new physical family

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after a 12-s warmup. This startup estimate assumes an intervention-free prefix.

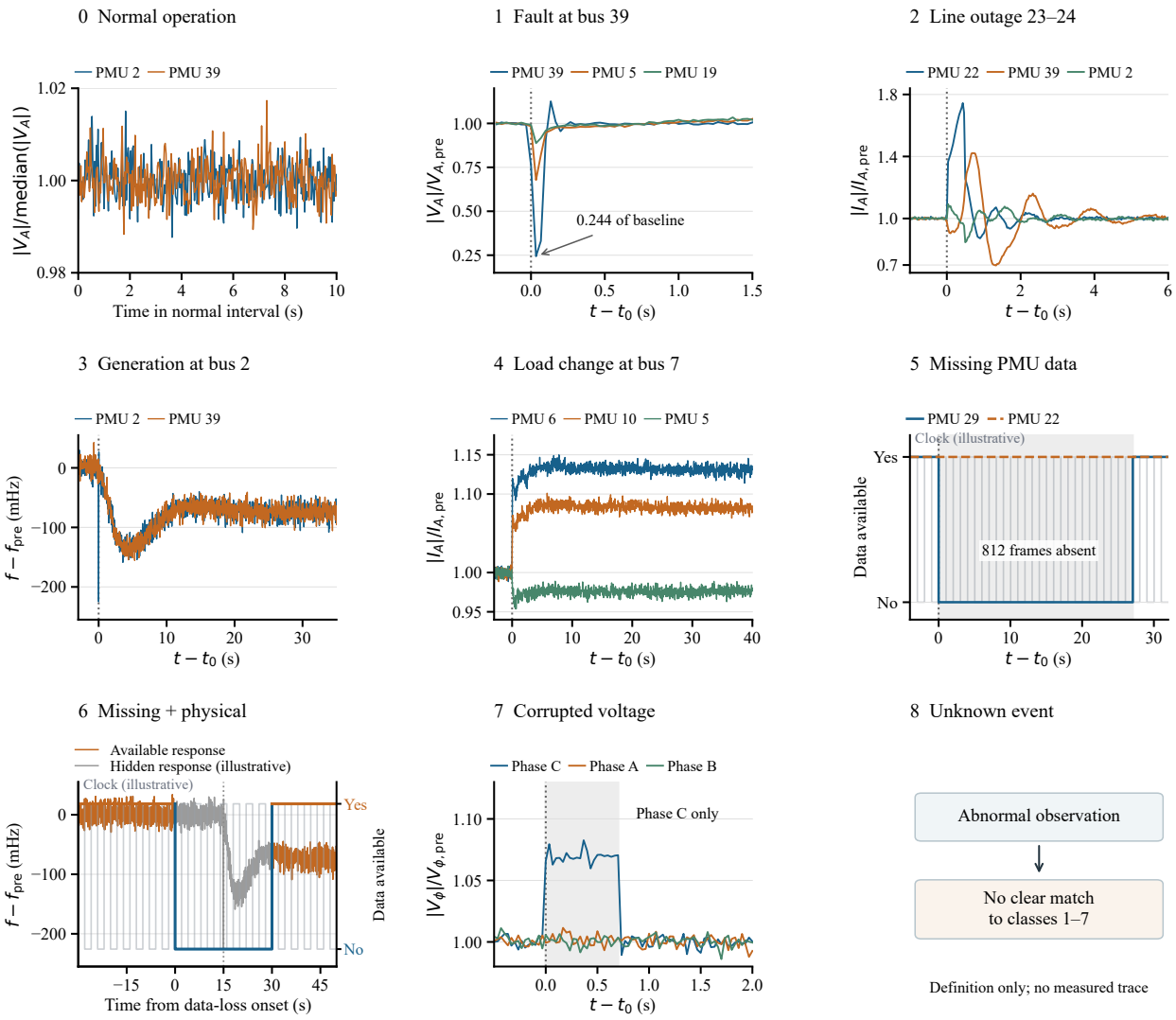


FIGURE 2. **Representative signatures from the archived RAW0001 reference simulation of the SGSMA 2026 competition, arranged by label 0–8. Class 0 shows a normal interval; classes 1–5 and 7 retain the original trace values. Class 6 applies an illustrative data-loss mask from 0 to 30 s to the PMU 2 frequency trace (generation onset at 15 s): orange shows the available response and Yes level; blue marks No, while gray shows the response hidden during the gap, not a detector reconstruction. The mask is not the recorded outage interval. The gray clock trains in panels 5 and 6 are illustrative, not measured report timing. Class 8 is a definition schematic, not an observed trace or a validated bank output. Event traces are aligned to source labels for illustration only. Only load and line mechanisms enter the main causal evaluation.**

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

At each report, the bank compares predictions from candidate intervention histories (Fig. 1(b)). Reduced state-space models carry the physical and instrumental state forward; their dynamics and measurement maps come from the network DAE. This avoids solving a full nonlinear simulation for every candidate. Conditional filters account for uncertain state and persistent model discrepancy, and the resulting evidence updates the weights of the competing histories.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology q, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with

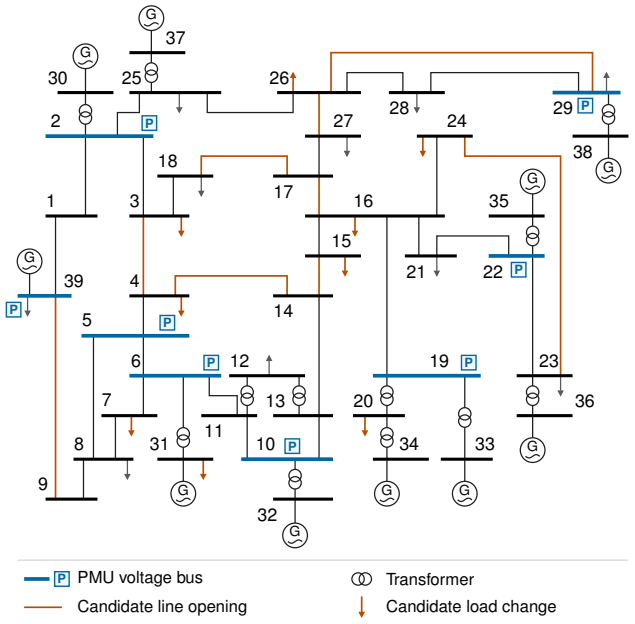


FIGURE 3. IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract.

a fixed angle reference, the derivatives at that consistent point give, suppressing ℓ,

$$\begin{aligned}\delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u),\\ \delta\dot{x} &= c_q + A_q^{red}\delta x + B_q^{red}\delta u,\\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Branch stamps in g_q and H_q encode network connectivity; eliminating unmeasured algebraic variables generally produces dense reduced operators. A line switch changes both the voltage response and the terminal operator in H_q. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant. The supplement checks these derivatives against nonlinear trajectories and tests a relinearized physical predictor separately from the history bank.

Exponential discretization integrates c_q over each mode interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned}\xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k,\\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining observation noise after instrument whitening. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section ''Implemented Predictor and Conditional Uncertainty'' gives the affine construction and its code correspondence. The network enters through these electrical operators, not a learned graph embedding. Updating conditional states and history weights does not adapt the physical parameters or measurement-noise model online. The supplement verifies a rotor-energy balance and its coupling through electrical graph ports. An invertible energy-coordinate representation preserves the physical predictions; energy reconstructed from the same PMU history is not scored as independent evidence.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S.

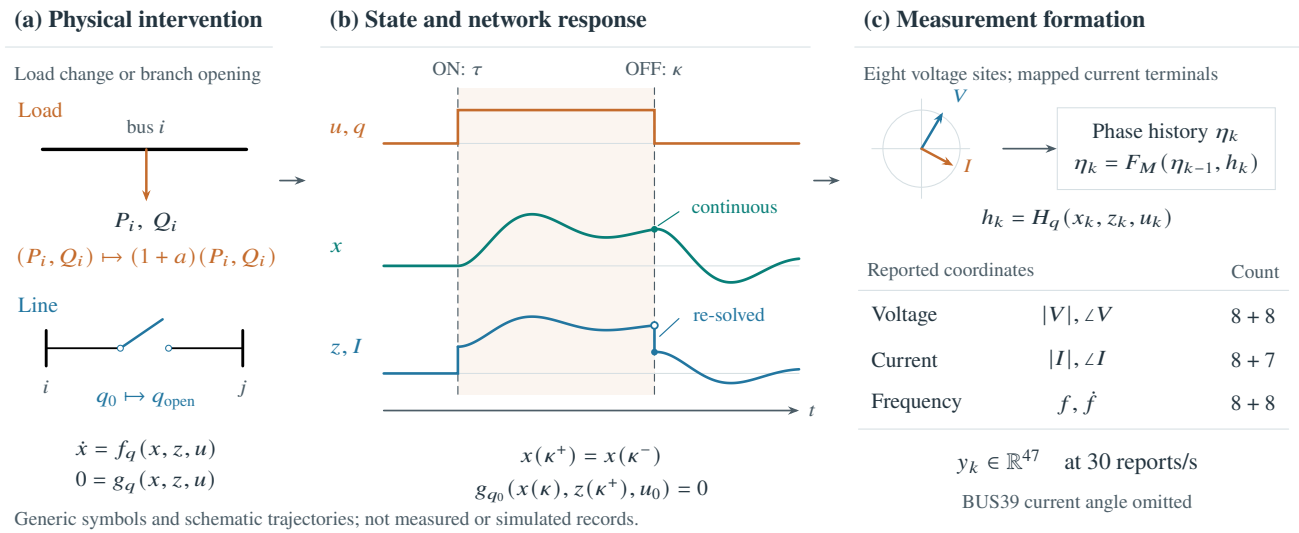


FIGURE 4. Illustrative chain from load/line intervention to causal PMU report. At OFF, the operator returns to baseline while x and phase-history η persist; algebraic voltages and currents are re-solved. Eight PMUs yield 47 coordinates at 30 reports/s; BUS39 current angle is omitted. Symbols and traces are schematic. Phase B/C are reconstructed from A; raw waveforms and harmonics are not inference inputs.

Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = {} & \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ & + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

As in separate-bias filtering [10], the conditional state mean is represented as $m_h + d_h a$; d_h is the Willsky–Jones signature of a unit severity propagated through the filter [4]. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter v_h. The affine conditional representation retains state–severity dependence:

$$\begin{aligned} \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ \mathrm{Var}(\xi \mid h) &= P_\rho + d_h V_{a,h} d_h^\top. \end{aligned} \tag{14}$$

Conditioning on past reports is implicit. The shared P_ρ solves the nominal-operator Riccati equation of regime ρ; S_o and K_o are then formed for each switched operator o. This is a steady-state approximation, not an exact covariance recursion along each switched history. At closure, the filter retains the augmented-state mixture mean and records the trace of the discarded amplitude and between-history covariance; it does not propagate that covariance. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight is updated with the one-step predictive density

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}, \tag{15}$$

$$p_\ell(y_k \mid \mathcal{F}_{k-1}) = \sum_{h'} \pi_{\ell,h',k-1} \sum_{h \in \mathcal{T}(h')} p_\ell(h \mid h')\, L_{\ell,h,k}. \tag{16}$$

The transition set $\mathcal{T}(h')$ contains continuation, restoration at either half-report phase for an eligible active history, and a

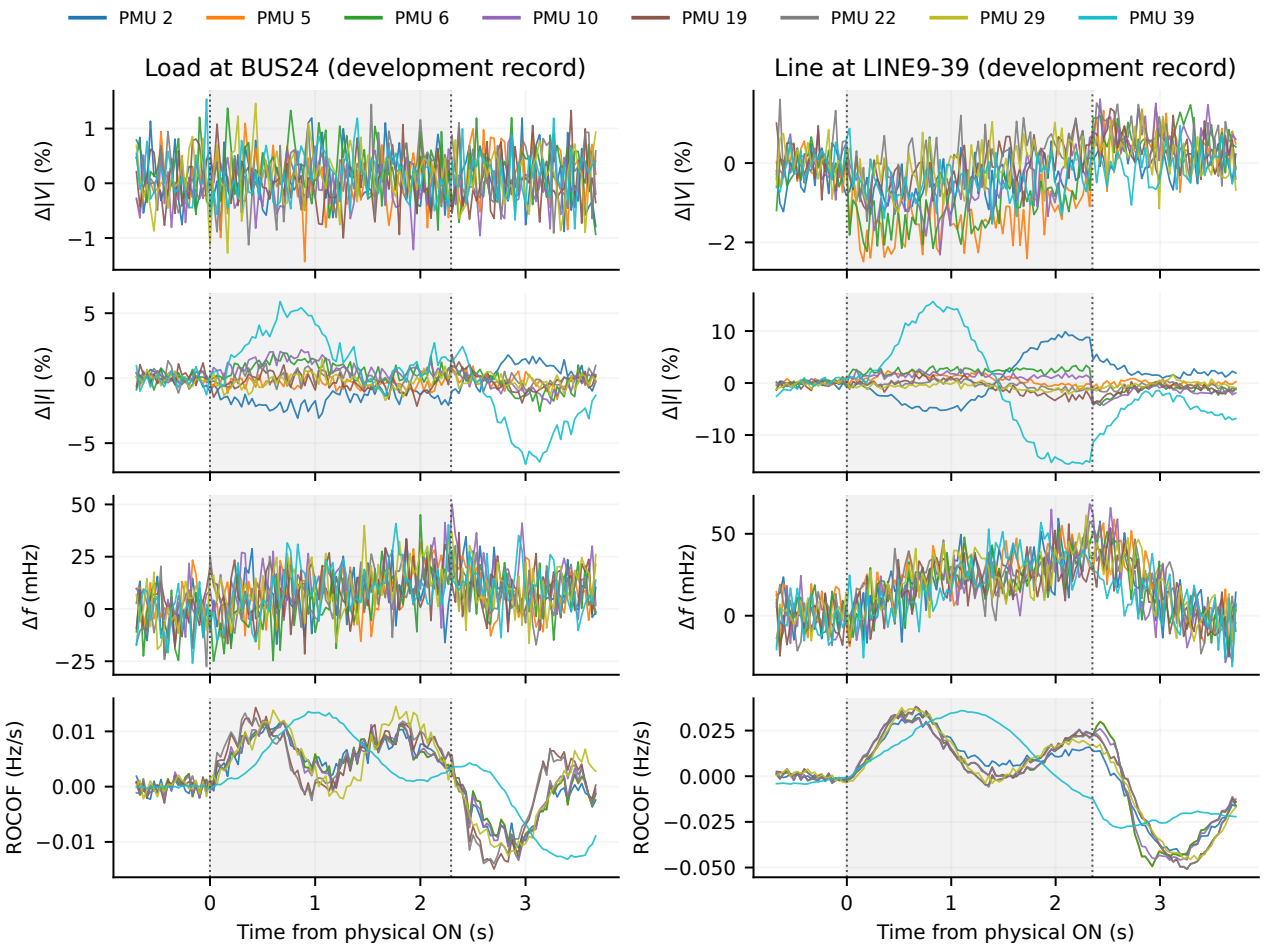


FIGURE 5. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.

birth of every catalog cell and severity interval at either phase from the no-intervention history; the transition probabilities sum to one, which the implementation asserts at every report. The density is evaluated before any state is corrected and before retrospective proposals are selected. Onsets proposed later for earlier reports change the inner history posterior through suffix replay but do not re-weight w_ℓ; this is the one point where the outer and inner recursions differ from a single joint posterior. Startup-fitting reports are not scored again. The mixture covers a finite set of network alternatives rather than a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{17}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{18}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

The implementation uses M for active-event publication and Γ for source diagnosis. Their update weights are ap-

proximate because of pruning and shared steady-state covariance approximations; they are not calibrated posterior probabilities. A descriptive algebraic decomposition of ΔM into within-model evolution and network-model reweighting is given in the supplement; it does not enter the estimator's decision rule.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

An alarm is published when the active mass $\sum_c M_{c,k}$ exceeds 0.99 for six consecutive reports. The rule has the structure of a Shiryaev change-time threshold under the geometric prior [9], but pruning and covariance approximations make 0.99 an operational threshold on approximate mass, not a calibrated posterior-probability guarantee. Six-report confirmation suppresses brief excursions. OFF is published when the restoration mass exceeds 0.99 for 15 reports. The diagnosis reports the family with the largest cause mass and its highest-mass source; episode-replacement rules handle a different source that begins before the previous episode closes. Every message records its emission time and the last report used, so a diagnosis can be correct but too late to precede physical OFF.

These are the development publication settings. The common-budget comparison in Section VII-A selects thresholds separately on calibration records. The nominal comparator is the same procedure with only the nominal network hypothesis.

The per-report birth and restoration hazards are $1-\exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$, fixed during method development before the independent schedules. They are deliberately conservative: test interventions last 2–5 s and are separated by 3–8 s, much shorter than the prior means of 60 and 30 s. A low birth hazard raises the evidence needed for an alarm by $-\log\pi_c$; Section V compares this margin with nominal signature energy. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The supplement maps each operation to its implementation and lists the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \;\propto\; L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h')\pi_{\ell,h',k-1}. \tag{19}$$

This is the mode-sequence update of IMM and M3H filtering [11], [12]. Three features are specific to the present problem: the modes are oriented electrical intervention operators derived from the network DAE, histories carry a restoration time so that a restored intervention remains a competing cause, and the output separates active intervention from originating cause through (18). Retrospective births reuse a stored pre-onset state, and confirmed closure compresses histories into the retained augmented-state mean.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

No event truth is available to the algorithm. Retrospective replay changes the inner history weights but does not re-weight the outer model densities. The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. Decisions use only reports already received. Computation time is reported separately in Section VII.

V. NOMINAL SIGNATURES AND STATE CONTINUITY

The local operators give a nominal signature for each candidate. We use these signatures to explain separation and the effect of resetting the physical state under the assumed model. The numerical comparisons with completed runs are retrospective.

A. SIGNATURES AND INFORMATION

Let intervention i start at report 0 with severity a and let the plant follow the local switched model (8) without noise. Run the conditional filter of candidate history j, started at the same onset, on these reports. Its innovation $e_j(k)$ and severity sensitivity $s_j(k)$ follow the recursion (13). The filter of the true history at the true severity has zero deterministic innovation in this construction. We measure nominal separation by a profiled signature energy,

$$\Lambda_{ij}(k) = \tfrac{1}{2} \min_{b\in[l,u]} \sum_{r=0}^{k} \left\| e_j(r) - s_j(r)\, b \right\|^2_{S_j^{-1}}, \tag{20}$$

where a line candidate has no severity to minimize over. The quantity Λ_{i0} compares against the no-intervention path; $\min_{j\neq i}\Lambda_{ij}$ measures separation from the nearest alternative at the supplied onset. These are deterministic diagnostics. For fixed Gaussian paths with means m_i, m_j and joint report covariances Σ_i, Σ_j, the expected log likelihood ratio under i is

$$\begin{aligned} D_{ij} = \tfrac{1}{2}\big[&(m_i - m_j)^\top \Sigma_j^{-1}(m_i - m_j) \\ &+ \operatorname{tr}(\Sigma_j^{-1}\Sigma_i) - d + \log(|\Sigma_j|/|\Sigma_i|)\big], \end{aligned} \tag{21}$$

Algorithm 1 Causal physical-history attribution

Require: Fixed plant/catalog, configuration, and PMU reports as they arrive
Ensure: Time-stamped detection, source diagnosis, and restoration messages
Fit operating context, reference, and noise from the permitted prefix
Initialize model/history states, weights, and pre-onset state store
for each newly received report y_k after startup **do**
  Map y_k to inference coordinates r_k
  Predict each retained history; carry physical state through OFF
  Save outer-model predictive densities before screening
  Propose eligible births from stored pre-onset states
  for each accepted proposal **do**
    Replay its suffix from the saved state, replacing old suffix scores
  end for
  Score histories before state correction; integrate load severity
  Apply line operators; correct states and update history weights
  Prune the beam; record discarded probability mass
  Update outer-model weights from the saved predictive densities
  Aggregate active-event mass M_k and cause mass Γ_k
  Apply consecutive-report alarm, diagnosis, and OFF rules
  if OFF is confirmed **then**
    Compress closed histories into the retained augmented-state mean
  end if
  Store current states and emit messages with the last report used
end for

where d is the stacked report dimension. The covariance terms cancel when the covariances coincide. Equation (20) profiles severity on deterministic signatures using the candidate innovation covariance; it is not, in general, the full Gaussian KL divergence above. Optimizing a deterministic signature, profiling severity from noisy observations and integrating it under a prior are different operations.

For an idealized comparison of one onset and source against the null,

$$\lambda^\star = \log 99 - \log \pi_c. \tag{22}$$

This is the score required for odds of 99 to 1 if π_c is used as the approximate prior odds. For a load with $\Lambda_{i0} = a^2 G_i/2$ and covariance scaled by α^2, the corresponding reference amplitude at horizon T is

$$a_{\mathrm{ref}}(\alpha, T) = \alpha\sqrt{2\lambda^\star / G_i(T)}. \tag{23}$$

This does not specify a detection probability or a threshold guarantee for the bank: unknown onset, competing paths, marginal likelihoods and consecutive-report confirmation all affect publication.

B. THE COST OF FORGETTING THE PHYSICAL STATE

Consider a noise-free, matched-model continuation with no new intervention after an inferred closure at report k_c. Suppose the conditional mean at closure equals the augmented state. Carrying this state gives zero deterministic innovation. If the physical coordinates are set to zero while the instrument and discrepancy states are kept, the no-intervention filter receives the deterministic bias

$$\bar{e}_0(k) = C_0 \Phi_0(k, k_c)\, \delta x(k_c) - \sum_{r=k_c}^{k-1} C_0 \Phi_0(k, r+1) K_0\, \bar{e}_0(r), \tag{24}$$

where $\delta x(k_c)$ is the zeroed physical part of the augmented state, $\Phi_0(k, r) = T_0^{k-r}$ and $C_0 = J_0$ are the no-intervention operators of (13), and K_0 is its gain. This is the free recovery response (17) seen through a filter that tries to absorb it. Any birth hypothesis j that explains part of this bias has a profiled energy gain $\Gamma_j^{sp}(k) = \Lambda_0^{reset}(k) - \Lambda_j^{reset}(k)$, where both energies use the same reports from k_c to k. A gain above $\lambda^\star$ crosses the idealized score reference; it does not guarantee a published alarm. Under the same construction with carry, the null energy is zero, so every alternative has gain $\Gamma_j^{sp,carry}(k) \le 0$. The positive part of the best gain is therefore zero. Reset bias depends on the excited modes, their decay, and their projection onto the observed channels.

C. RETROSPECTIVE DIAGNOSTICS

We evaluated the signature and reset diagnostics in the nominal quiet covariance regime. The true-path deterministic residual was below 10^{-8} nats. Fig. 6 summarizes separation under those conditions.

For a 10% load step, the one-second nominal separation from the null ranges from 4603 to 6.9×10^6 nats; for lines it is at least 4.9×10^6 nats. These large margins are consistent with the short delays under matched conditions. The reference amplitude under uniform noise scaling is highest at bus 31. Neither diagnostic accounts for independent per-PMU noise or a shifted plant. The reset diagnostic (Section VII-E1) evaluates the deterministic effect of discarding a previously estimated state.

A separate channel diagnostic holds source amplitudes and a common temporal covariance fixed. At 1 s, adding terminal currents to voltage increases the minimum pairwise separation by a median factor of 1.35 across the 18 sources (range 1.03–8.41). Allowing rival loads to fit severity over the admitted interval gives a median factor of 1.53. The supplement derives both diagnostics and shows how fitting severity can reduce source separation. Neither result measures attribution accuracy.

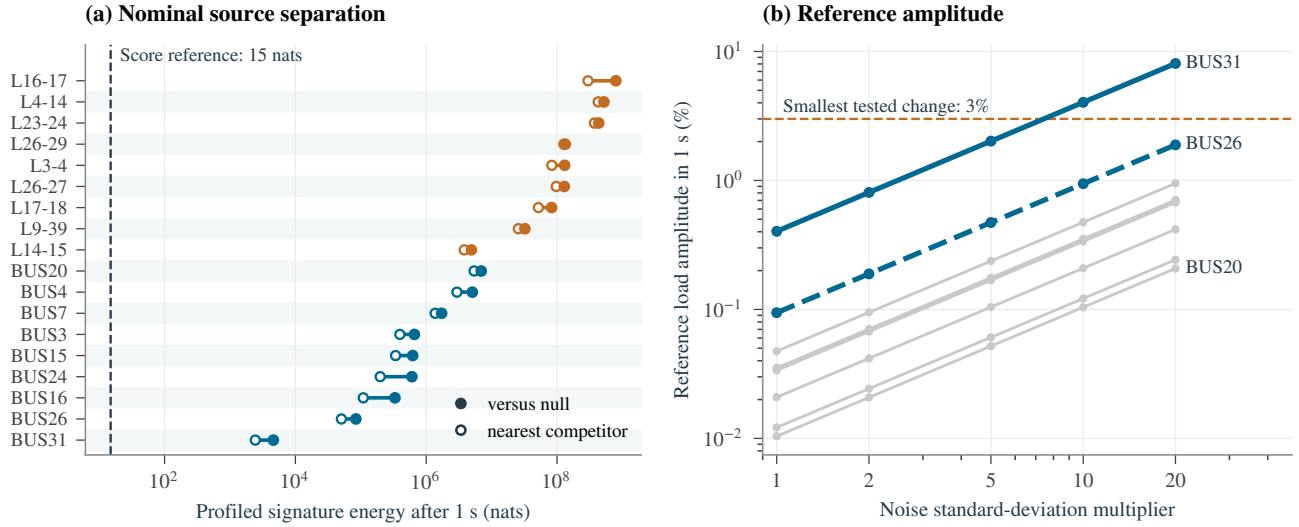


FIGURE 6. **Retrospective signature diagnostics using the nominal switched-DAE operators. (a) Profiled signature energy after 1 s against the no-intervention history (filled circles) and against the nearest competing source (open circles), for a 10% load step or a line opening; the dashed line marks the idealized score reference $\lambda^\star$. (b) Reference load amplitude at 1 s, (23), versus a uniform multiplier of the calibrated noise; gray lines are the other seven load buses.**

VI. EXPERIMENTAL DESIGN

The primary comparison measures timely source attribution and extra alarms for the history bank, a continuous GLR and ExtraTrees after calibrating each to the same normal-alarm budget. A separate evaluation varies network parameters and noise, while component controls examine state reset and severity scoring. Table 4 distinguishes these populations; repeated conditions of a parent sequence are paired observations.

The plant is an IEEE-39 model in Julia/PowerDynamics with Sauer–Pai machines, IEEE Type I exciters, TGOV1 governors and constant-impedance ZIP loads. Each physical parent is integrated continuously, preserving differential states through ON/OFF. The PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001. At a switch-aligned report the export uses the post-switch algebraic value. The bus-31 candidate is the existing 9.2-MW/4.6-Mvar load at the generator bus that serves as slack only in the initial power flow; the dynamic model has no slack bus, so its load change is absorbed by machine dynamics like any other.

Instrument noise follows a stochastic model fitted to normal segments of the competition reference record: per-family vector autoregressions with a strong common factor across PMUs. Voltage-magnitude noise is about 0.4% of nominal, but the noise of angle *differences* between PMUs is only 10^{-4} to 10^{-2} degrees, because most angle noise is common to all units. This small differential noise is what makes megawatt-scale changes observable at remote PMUs (Section V); it is a property of the reference data, smaller than the idiosyncratic error expected from field PMUs [48]. Section VII-F therefore tests larger and independent per-PMU noise.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

TABLE 3. **Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.**

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

The five-model bank and its nominal version were fixed before the independent schedules were drawn. Both ExtraTrees models were trained specifically for the causal load/line catalog, using 32 earlier physical parents for training and eight for development validation. Both training sets include normal operation and all 18 candidate sources. The reference uses 846 causal features from 1-, 5- and 10-s windows; the short-window variant adds 611 features from 3-, 8- and 15-report windows. Each model has 400 trees with the same predeclared hyperparameters. They inherit feature families from the SGSMA project, but their fitted weights are distinct from the original hackathon submission. These models and their adapters were fixed before the present independent schedules; no retraining uses the current evaluation records. All methods receive the same 30-fps observations. The supplementary reproduction table records the training lineage and model hashes.

The learned adapter publishes a change after six consec-

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Common-budget comparison	18 event parents, 108 interventions; 24 normal parents	Timely family/source, extra alarms and paired differences	Complete procedures; thresholds selected on separate calibration data
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

utive identical argmax labels; its tree votes are uncalibrated scores. In its own development validation the reference model already attributed no load before OFF and the short-window variant four, while lines were attributed well; both were nevertheless retained unchanged under a predeclared rule. They are therefore reported as fixed adapters at their original operating points, not as the best attainable learned baseline. Learners retrained on the bank's simulations locate at most 49 of 72 held-out events, against 72 for the bank (supplement).

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

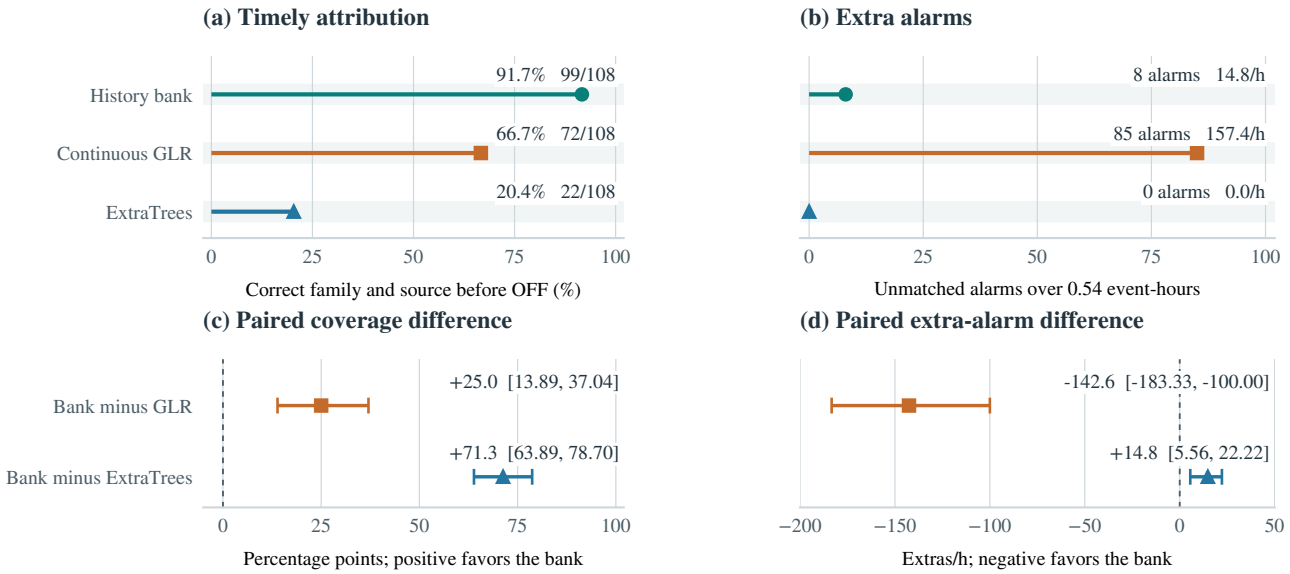
Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time.

VII. RESULTS AND CONTROLLED COMPARISONS

A. COMPARISON AT A COMMON CALIBRATION BUDGET

On a separate cohort, the nominal bank and the continuous generalized likelihood ratio (GLR) observer used the same nominal operators, eight PMUs, source catalog and permitted startup prefix. The GLR carries one committed state through restoration. ExtraTrees was refitted on disjoint simulated parents covering the same catalog. A common publication clock and calibration procedure selected a threshold for each arm under a normal-alarm budget of one per hour. The fixed methods were then scored once on 18 event parents and 24 normal parents. Interventions did not overlap, and the next onset was 3–8 s after physical restoration.

Figure 7 shows a 25.0 percentage point gain over GLR (paired 95% CI [13.89, 37.04]), with fewer extra alarms. The gain over ExtraTrees is 71.3 points (95% CI [63.89, 78.70]), with more extra alarms. The intervals resample entire physical parents within the network/noise strata; events within a parent do not count as independent samples. All three methods



Normal records: 0 alarms in 3.92 h per method; one-sided 95% Poisson upper bound 0.76/h.
Paired 95% intervals: 10,000 resamples of 18 event parents. Each method meets the calibration alarm budget.

FIGURE 7. Coverage and alarm cost on held-out sequences. (a) Correct family and source before OFF, out of 108 interventions. (b) Unmatched alarms over 0.54 monitored event-hours. (c,d) History bank minus each comparator, with 95% intervals from 10,000 paired resamples of 18 event parents. Positive coverage differences and negative alarm-rate differences favor the bank. The bank improves both outcomes over GLR but trades more extras for higher coverage than ExtraTrees. Thresholds were selected on separate calibration data under a target of one normal alarm/h; normal exposure and its Poisson upper bound are shown separately.

had zero normal alarms in 3.92 monitored hours per method (one-sided 95% Poisson upper bound: 0.76/h). The result is conditional on the count model and the aggregate exposure. The two physical methods differ in candidate search, score semantics and commitment, so the comparison does not isolate the effect of explicit recovery histories.

B. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched alarms on event records, and its nominal counterpart 643/648 with 92. These are paired replays of 108 interventions in 18 physical parents, not 648 independent events. On the 72 normal records (42,336 s, 11.8 h of monitored time per method) neither bank raises an alarm, which bounds the normal false-alarm rate below 0.26 per hour at 95% confidence (Poisson). Every matched diagnosis emitted before OFF names the correct family and source; the timely misses are late or absent diagnoses. Table 5 and Fig. 8 give the family split.

The extra alarms concentrate where the network model is wrong: 68 of the 72 occur when the true R/X perturbation lies outside the five-model bank, 68 alarms in 1.08 h of out-of-bank monitored event-record time (about 63 per hour), against four in the other four strata together. Out-of-bank mismatch also produces the only four timely misses. The supplement gives the paired parent-cluster intervals; the physical bank's median alarm delay is about 0.27 s in the matched and in-bank conditions, comparable to the learned adapters (0.19–0.30 s).

TABLE 5. Held-out continuous-stream decisions. Each family has 324 replays of 54 distinct interventions; extras are unmatched alarms on event records. The ExtraTrees adapters keep their original operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

We compare these delays retrospectively with the nominal signature crossing. For the nominal bank on matched networks, line alarms arrive after a median 0.18 s, at the six-report confirmation floor of 0.20 s less the sub-report onset phase, and doubling the noise leaves them unchanged. Load alarms arrive after 0.35 s, again unchanged by doubled noise but about 4 reports after the nominal signature reference crossing. Line delays are consistent with the publication floor, while load delays exceed that model-based reference. The two ExtraTrees adapters recover lines (315 and 322 of 324) but few loads (15 and 30). The diagnosis-time distributions in the supplement show that this is a coverage failure: every load diagnosis the adapters do emit before OFF is correct, but most load interventions receive none. A per-report audit on four further matched records (12 load and 12 line interventions, 1,245 and 1,143 in-event reports) locates the failure before any publication rule. During every load intervention the ref-

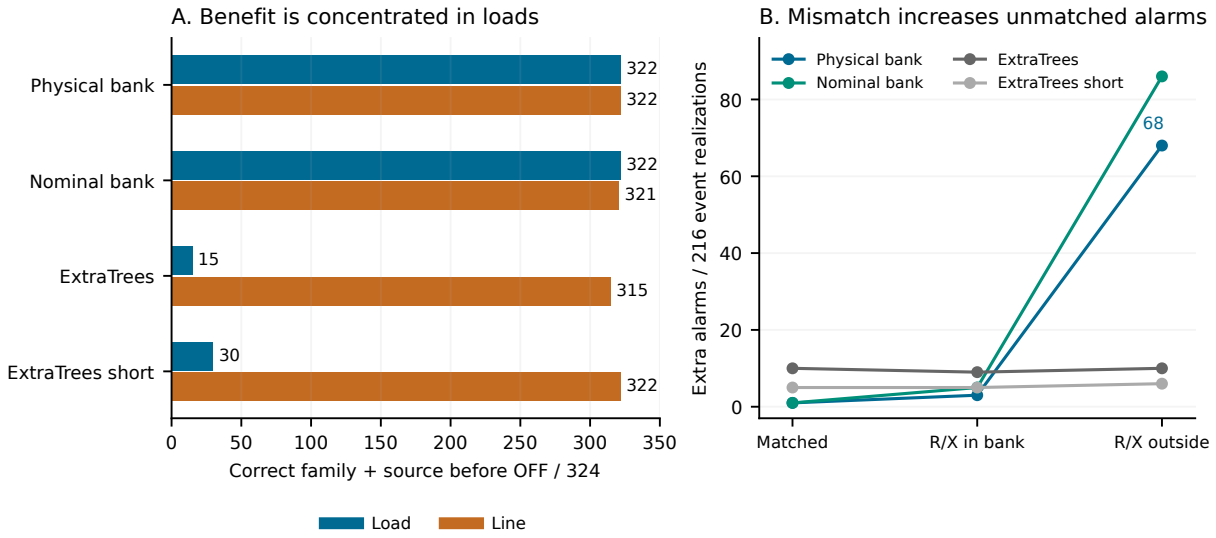


FIGURE 8. **Held-out continuous evaluation. (A) Timely correct family and source by event family, with 324 paired replays per family. (B) Unmatched alarms on event records for the two noise levels of each R/X condition (216 event replays per condition). Each method keeps its own publication rule.**

erence model's per-report argmax is the normal class in all reports, and the median of the largest true-class score is 0.09; during line interventions the true line wins 96% of the reports with median peak score 0.93. The short-window variant ranks the true load first in only 2 of 12 load interventions. The learned representation thus does not separate 3–14% load changes from normal operation, whereas the nominal physical signatures have large separation energies for the same changes (Fig. 6). The same pattern was already present in their development validation (Section VI), so the comparison shows that the physical bank recovers load interventions that these particular adapters miss; it is not a test of the best attainable learned detector at a common alarm budget. In the persistent-line test, both banks and both adapters identify the correct line first on all 162 openings. The adapter of Chen *et al.*'s quickest-change detector identifies 74/162; 66 records stop before the opening and 22 afterward on a wrong line, and it stops on all 72 paired normal records (first stop after 2,522 s of cumulative normal exposure, against 42,336 s monitored without alarm by the other methods). Its white-injection, quasi-static model does not describe the colored, dynamically filtered measurements of this plant, so its threshold is far from its nominal false-alarm rate. We therefore read this result as a test of transfer under model mismatch, not as a ranking of quickest-change detection. After the first identification, the two learned adapters raise 164 and 179 further unmatched alarms on the same persistent records, against 0 for the physical bank. Stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

Table 6 restates the evaluation in the metrics prescribed by the competition guide. Per-report detection F1 is 0.91 for the bank and 0.58–0.64 for the learned adapters, whose precision is higher because they alarm only on the line interventions. None of the methods raises an alarm on the normal records. The bank has no trained weights: its parameters are the network model, the noise model estimated from normal data, and a few prior and publication constants fixed during development. The nominal bank processes one minute of PMU data in 48 s on one CPU thread.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A development trajectory, shown in the supplement, illustrates the timing criterion. The network keeps moving after OFF, and a later event is evaluated against that predicted response. Under out-of-bank network error, the bus-31 source is identified only after its restoration: correct, but not timely. The active-mass trace, the ΔM decomposition and the development counts are in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

A development control keeps one committed augmented physical state through restoration instead of competing histories. It uses the same nominal plant and measurement operators, profiles severity over short alternatives of at most 19 reports and applies the same six-report publication clock. On 54 development replays (three parents, three R/X conditions), this single-state generalized likelihood ratio (GLR) detector attains 37/54 timely decisions with 37 extras, against 44/54 and 6 for the nominal history bank and 51/54 and 3 for the five-model bank. The development schedules pack six interventions into 80-s records instead of 120 s, which likely explains why the nominal bank scores lower here than in Table 5. The three procedures also differ in onset search and severity scoring, so this comparison indicates, but does not isolate, the benefit of explicit recovery histories over a single continuous state.

TABLE 6. Independent evaluation in the metric vocabulary of the competition guide (648 interventions, 706 min of normal records, three R/X conditions pooled). Detection is scored per report; Top-k requires the correct family and source among the first k candidates. Run time is per minute of PMU data on one CPU thread.

Method	Precision	Recall	F1	FA/min	Delay (s)	Top-1	Top-3	Run time (s/min)
Physical bank, five R/X models	0.888	0.926	0.907	0	0.20	0.995	0.995	179
Physical bank, nominal model	0.874	0.927	0.900	0	0.27	0.995	0.995	48
Submitted ExtraTrees	0.938	0.424	0.584	0	0.27	0.509	0.509	–
Retrained short-window ExtraTrees	0.944	0.483	0.639	0	0.19	0.543	0.543	–

E. SEPARATING SEVERITY SCORING FROM STATE CONTINUITY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{25}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{26}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 9 report coverage and alarm cost for the four arms. With the state carried, marginal and bounded-profile severity scores tie on every reported endpoint. Resetting the 114 estimated physical-state deviations at inferred closure reduces timely attribution from 108/108 to 75/108 and raises the unmatched alarms from 1 to 166 under both scores. The reset leaves instrumental state, covariance regime, prior weights, thresholds and publication rules intact, and the simulated plant never resets.

Fig. 10 compares this outcome with the retrospective reset diagnostic. The profiled energy gain exceeds $\lambda^\star$ after every intervention, by at least two and up to seven orders of magnitude, and the reset arm is harmed after 84 of the 108 interventions: an extra alarm in the following gap or a lost timely attribution of the next one. Harm occurs after 51/54 line events and 33/54 load events. The bus-31 events carry the smallest profiled gain and are harmed after three of seven. The diagnostic shows that reset leaves a residual that event candidates can explain. Its magnitude alone does not determine the observed outcome: episode handling and publication rules also matter. These are within-simulator contrasts at the original thresholds; the two scores are compared at equal thresholds, not at equal alarm rates.

F. SENSITIVITY TO INDEPENDENT PMU NOISE

To examine sensitivity beyond the calibrated noise, we reran the nominal bank with the same equations and publication thresholds on four held-out calibration event records (12 load and 12 line interventions, matched network) and two 600-s normal records (0.33 h). Six noise scenarios were generated with the simulator's own noise stream: the original realization scaled by $\times 1$, $\times 5$, $\times 10$ and $\times 20$, and the original plus independent per-PMU relative-magnitude and angle noise, each with standard deviation $\sigma = 0.0005$ or 0.001 (angles in radians). The added perturbation has approximately 0.071% or 0.141% RMS total vector error (TVE), respectively, before combination with the original noise. Voltage-angle noise is propagated to frequency and ROCOF by the simulator's 21-report estimator. The $\times 1$ scenario reproduces the original predictions message for message. The bank re-estimates its noise scale and observation discrepancy from the noisy 300-report prefix; the operating point is kept at its nominal-noise estimate because derivative matrices exist only there. The implementation refuses to start when the estimated noise scale leaves its calibrated support of $[0.1, 10]$; for this test

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

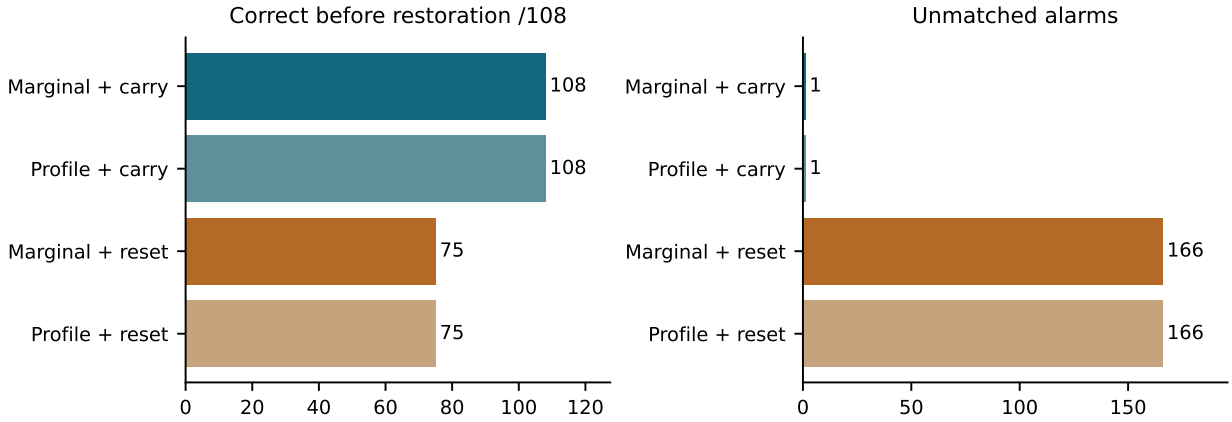


FIGURE 9. Severity-scoring and state-carry study on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the alarm rate may differ between arms.

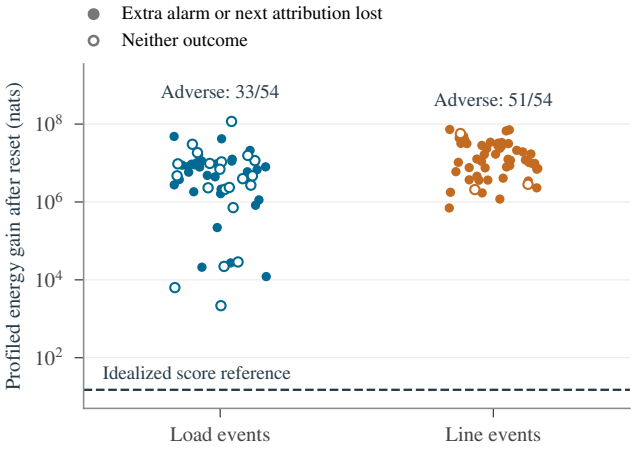


FIGURE 10. Retrospective profiled energy gain after a state reset, (24), for the 108 interventions of the controlled study. Filled: an extra alarm in the following gap or a lost timely attribution of the next intervention; open: neither. The dashed line is the idealized score reference, not the publication threshold. Under matched noise-free continuation, carrying the state gives no positive gain over the null.

only, that bound was widened, and every other line of code is unchanged.

Scaling the calibrated noise by up to twenty preserves all line attributions in this auxiliary cohort. One load intervention, a 4.1% change at bus 26, is missed at $\times 5$ and $\times 10$ but not at $\times 20$. Its amplitude exceeds the nominal reference at all three levels. That disagreement illustrates why a signature-energy crossing cannot guarantee a published attribution. The median load alarm delay changes from 0.35 to 0.36 s. No scenario produces an extra alarm on event records or an alarm on the normal records.

Independent per-PMU noise has a larger effect: the bank attributes 3/12 load interventions at $\sigma = 0.0005$ and 0/12 at $\sigma = 0.001$, while the corresponding line counts are 12/12 and 10/12. A matched-signature diagnostic using only the added independent noise still assigns substantial energy to these load responses (supplement). That restricted diagnostic excludes source search and the bank's covariance approximation; it does not establish achievable attribution accuracy. The bank fits one noise scale per signal family while retaining the reference cross-PMU correlation. Independent noise raises the fitted angle-noise scale and weakens load evidence under that model. Estimating cross-PMU covariance is therefore a candidate remedy that requires a separate controlled evaluation. These results do not establish readiness for field PMUs.

G. COMPUTATIONAL COST

Table 8 records the stages of the evaluation campaign, which ran the five-model bank on one thread. Its history inference takes a median 358.7 s per 120-s record (99.6 ms per report), three times slower than the 30-report/s input, so the five-model bank as implemented is an offline causal replay. The cost is dominated by the number of network hypotheses. Re-running the unchanged nominal bank alone on the same hardware and thread settings (Section VII-F) gives 26.4 ms per report (median over 6 records, maximum 30.0 ms), below the 33.3-ms report interval; the nominal bank attributed 643/648 interventions in the independent evaluation. These

TABLE 8. Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

are amortized processing rates on a desktop CPU; per-report latency percentiles and memory were not instrumented, and the 80-s startup bank construction is a one-time cost. Decision timestamps elsewhere in the paper are report times, not wall-clock latency.

VIII. DISCUSSION

A. WHAT STATE CONTINUITY EXPLAINS

A restored operator can leave a nonzero physical state, so returning the predictor to equilibrium introduces a residual that another candidate can explain. The reset control shows this failure at fixed thresholds; bounded profiling and marginalization tie in that control. The nominal signature diagnostics help explain short delays under the assumed noise model. They do not predict operational error rates. The bank has no online applicability gate to reject decisions after a plant change invalidates its models. Sparse-PMU transient-stability work has reported an explicit self-check for that applicability question [35]. The present signature diagnostics were evaluated retrospectively and do not suppress alarms.

B. COVERAGE, ALARM COST AND SCOPE

In the held-out comparison, the history-bank procedure had higher timely coverage and fewer event-record extra alarms than the continuous GLR. ExtraTrees emits fewer extra alarms but misses more interventions. These are contrasts between complete procedures calibrated to a common normal-alarm target; zero observed normal alarms give upper bounds, not equality of the underlying rates. Under an adequate Markov model, the full augmented-state distribution is sufficient for prediction. Compressing histories to their mean can discard covariance and distinct state alternatives. A history-bank advantage over that compression would therefore concern a practical representation of uncertainty, rather than information unavailable to every state observer. The supplementary distributional analysis separates predictive sufficiency from historical cause attribution. The present comparison does not isolate these effects.

C. TRANSFER AND MEASUREMENT LIMITS

Observations and predictors use the same simulator family. The catalog is fixed, the gaps between interventions are 3–8 s, and no intervention overlaps another. Transfer of the dynamic bank to a distinct plant, shorter gaps and unseen sources are untested. Under independent per-PMU noise at 0.071% added RMS TVE, the bank misses most load changes. This limits transfer beyond the calibrated measurement model. The terminal map retains the provisional bus-5 current terminal and excludes the bus-39 current angle. A dropout with held observations prevents recovery in the auxiliary stress test. The bank did not start on the retrospective competition record; the integrity checks and separate attributor in the supplement do not validate the dynamic bank on that record.

The tested IEEE-39 plant also has no grid-forming inverter controller dynamics. Applying the method to converter-dominated networks would require adding the relevant control and current-limiting states and testing those predictors anew; GFMI surveys describe several such modeling choices [53].

D. APPROXIMATION AND PROCESSING COST

Pruning, steady-state covariances, startup fitting and compression make the attribution weights approximate. Retrospective proposals change inner history weights without updating the outer model weights. The nominal bank processed reports faster than they arrived in a measured replay on one thread; the five-model version did not. This timing is specific to the reported hardware and records.

E. ADAPTATION WITHOUT ABSORBING AN INTERVENTION

Adaptive noise estimation has already been studied on IEEE-39 [54]. Supplementary nominal-operator diagnostics support retaining recovery covariance when onset and amplitude are known. However, continuous noise adaptation did not yield a conclusive attribution gain, and a quiet-history gate rarely admitted updates. Two arms exceeded the normal-alarm target on new records. These tests use a local likelihood scan, not the reported history bank; they motivate covariance propagation but do not validate adaptive event inference. The supplement specifies that extension and separates it from cyber analysis, which needs its own alerts and validation. Missing reports or physical residuals alone do not identify an attack.

IX. CONCLUSION

We formulated sequential load and line attribution from sparse PMUs as a comparison of switched physical histories. The predictor retains estimated state at restoration and separates an active intervention from a preceding cause of recovery. On a held-out simulation under a common calibration budget, the nominal bank made timely, correct family-and-source attributions for 99/108 interventions, versus 72/108 for a continuous physical GLR and 22/108 for ExtraTrees. The respective unmatched alarm counts were 8, 85 and 0. A fixed-threshold state-reset control shows the cost of discarding the estimated state. The complete-procedure comparison does not isolate the benefit of retaining recovery histories from the loss of uncertainty under compression. The findings apply to sequential interventions from a known catalog in one

simulator family. Transfer to another dynamic plant, shorter gaps and unseen sources remain untested; independent per-PMU noise at 0.071% added RMS TVE severely degrades load attribution.

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JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.



MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.



PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.



DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.